

MAT211: Calculus-II

Problem Sheet #01

Date: February 20, 2026

Topic: Function, Level sets, Limit

Extra Practice (Not for marks)

1. Which of the following subsets of $\mathbb{R}^n$ is open? closed? neither? Prove your answer.

- $\left\{ \begin{bmatrix} x \\ y \end{bmatrix} : y > 0 \right\} \subset \mathbb{R}^2$
- $\left\{ \begin{bmatrix} x \\ y \end{bmatrix} : y \geq 0 \right\} \subset \mathbb{R}^2$
- $\left\{ \begin{bmatrix} x \\ y \end{bmatrix} : y > x \right\} \subset \mathbb{R}^2$
- $\left\{ \begin{bmatrix} x \\ y \end{bmatrix} : xy \neq 0 \right\} \subset \mathbb{R}^2$

2. Identify $\mathcal{M}_{m \times n}$ the set of $m \times n$ matrices, with $\mathbb{R}^{mn}$ in the obvious way.

- Prove that when $n = 2$ or 3 , the set of $n \times n$ matrices with nonzero determinant is an open subset of $\mathcal{M}_{n \times n}$.
- Prove that the set of $n \times n$ matrices A satisfying $A^T A = I_n$ is a closed subset of $\mathcal{M}_{n \times n}$.

3. Sketch families of level curves and the graphs of the following functions f .

- $f\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = 1 - y$
- $f\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = x^2 - y^2$
- $f\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = x^2 - y$
- $f\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = xy$

4. Consider the surfaces

$$X = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x^2 + y^2 - z^2 = 1 \right\} \quad \text{and} \quad Y = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x^2 + y^2 - z^2 = -1 \right\}.$$

- Sketch the surfaces.
- **(Optional)** Give a rigorous argument (not merely based on your pictures) that every pair of points of X can be joined by a curve in X but that the same is not true of Y .

5. Use $\epsilon - \delta$ style statement to show the limit:

- $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x - 1} = 2$
- $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$